

3 CORE 200 Reading | Module 3 | Indeterministic Causation

Contents

3	CORE 200 Reading Module 3 Indeterministic Causation	1
3.0.1	Module 3 Introduction	1
3.0.2	Concepts Covered in Module 3	2
3.1	Exploration	2
3.1.1	Module 3 A Concrete Example: Cell Phones	2
3.1.2	Module 3 An Abstract Example: Red and Blue Squares	3
3.1.3	Module 3 Individuals and Populations	4
3.1.4	Module 3 Cell Phones Revisited	5
3.1.5	Module 3 Red and Blue Squares Revisited	5
3.2	Indeterminism and Pseudo-Indeterminism	6
3.2.1	Module 3 Introduction	6
3.2.2	Module 3 Pseudo-Indeterminism	6
3.2.3	Module 3 True Indeterminism	7
3.2.4	Module 3 The Practical Situation	8
3.3	Indeterministic Causation	8
3.3.1	Module 3 Introduction	8
3.3.2	Module 3 Cell Phones and Indeterminism	9
3.3.3	Module 3 Quantifying Indeterministic Causation	9
3.4	Defining Causation with Indeterminism	10
3.4.1	Module 3 Direct Causation with Indeterminism	10
3.4.2	Module 3 Direct vs. Indirect Causation	12
3.5	Summary	13
3.5.1	Module 3 Module Summary	13
3.5.2	Concepts Covered in Module 3	13

3.0.1 Module 3 | Introduction

Suppose you have agreed to buy a house for \$100,000, but you are still negotiating with the current owners about what needs to be fixed before they move out. As you walk through the house with the inspector you've hired, you notice that it has really old heaters called Taylor burners in really old fireplaces. The inspector says the old heaters need to come out because they have asbestos lining, and asbestos causes cancer. Mrs. Smith, the current owner, is offended and doesn't want to pay for professionals to remove the asbestos. "I raised five kids here, and it didn't do them any harm," she says. "My relatives lived in houses with these heaters for 40 years, and none of *them* got cancer," she continues. How should you reply? If asbestos really causes cancer, then why didn't her children or her relatives get it?

So far, all the causal relationships we have studied are "deterministic," that is, the effect is always completely determined by the values assigned to the causes. In an "indeterministic" causal relationship, the effect is not always determined by the causal assignments. For example, smoking causes cancer, but doesn't determine it. Some people who smoke live to 100 and never get lung cancer.

Almost every causal relationship we study in the social, behavioral, and medical sciences at least *appears* to be indeterministic. Having wealthy parents is a cause of wealth, but it does not guarantee it. Eating well and exercising regularly cause good health, but they don't guarantee it. We cannot be satisfied with an account of causation among variables in which causes must completely determine the value of their effects.

There are three core ideas in this module. The first is that causal relations are sometimes indeterministic. The second is that sometimes they are indeterministic in appearance only. In most cases we don't explicitly *mention* all of the causes that act on an effect of interest, but only some of them. If we did list *all* of the causes of a particular effect, then the relationship between the causes and the effect might well be deterministic. When we leave some out, however, the relationship will *appear* to be *indeterministic*.

The third core idea is this: even though a particular causal assignment might not always determine whether a given effect will or will not occur, it can determine the *chances*, or the *probability* that the effect will occur.



At the very least, switching from one causal assignment to another can change the chances that the effect will occur. Being exposed to asbestos, for example, does not guarantee that you will get cancer, but it does raise the *chances* that you will get it compared with not being exposed. Even though Mrs. Smith's children and relatives got lucky, you don't want to expose your children to asbestos, because to do so would increase their chances for getting cancer later in their life.

The idea that causes only determine the *probability* of their effects is the main reason why causal science needs statistics. If smoking had to lead to lung cancer in every single case before we agreed to say that smoking causes cancer, then a single anecdote about some old man who smoked until 99 but died of old age with no trace of lung cancer would be enough to defeat the causal claim and thus alter our policy of labeling cigarette packs with warnings like: "Smoking causes Cancer." We wouldn't need to keep data around about thousands of smokers or lung cancer patients. Saying that smoking changes the *chances* of lung cancer is another matter entirely. To learn about the effect of smoking on the chances of lung cancer, we need to know the chances of getting lung cancer after being made to smoke, and the chances of getting it if after being made not to smoke. To answer these questions we need to collect data. Whether or not the data we measure confirm that the chances of lung cancer are *really* different is the province of statistics.

The module that follows covers six concepts:

3.0.2 Concepts Covered in Module 3

1. Indeterministic Response Structures
2. Sources of Indeterminism
3. Pseudo-indeterminism vs. True Indeterminism
4. Quantifying Indeterministic Causation
5. Definition of Indeterministic Causation
6. Direct vs. Indirect Causation

3.1 Exploration

3.1.1 Module 3 | A Concrete Example: Cell Phones

We begin this module by illustrating the first core idea: that in cases where we don't explicitly mention all the causes of an effect, the causes we do mention do not appear to fully *determine* the effect. Later in the module we delve into how causes can change the *chances* of their effects. We leave all the statistics out – that comes much later.

Suppose we simulate on a computer a situation in which you are traveling in a car and you want to make several calls on your cell phone. You can tap the "SEND" button to place a call, and tap on the "END" button to end a call. The receiving phone on will ring if your call got through.

In real life, calls from cell phones in cars do not always go through. So this is clearly a case in which the cause, tapping on the "SEND" button, does not guarantee that the effect, the receiving phone ringing, will occur. Here, and in the next few sections, we abstract from this particular case to the general concept of indeterministic causation. First, we translate this case into the language of variables and response structures. This will allow us to discuss indeterministic causation via indeterministic response structures.

The simulated cell phone system, as we have presented it to you, includes two variables, each of which has two values:

CELL PHONE SYSTEM	
Variable	Values
BUTTON PUSHED	[Send, End]
CALL CONNECTED	[Yes, No]

Consider the effect: CALL CONNECTED. What is the *response structure* for this effect? The response structure for an effect lists the values that the effect takes on for every possible causal assignment of the other variables in the system. So to build the response structure, first enumerate all the possible causal assignments, and then fill in the value of the effect for each. In this case there are exactly two causal assignments, one for each value we might give to the Button Pushed variable:

RESPONSE STRUCTURE FOR THE CELL PHONE SYSTEM	
Causal Assignment	Effect: Call Connected
BUTTON PUSHED = Send	???
BUTTON PUSHED = End	No

Hitting the End button *does* determine the effect: the call is not connected. The problem is assigning a value to the effect for the first causal assignment, BUTTON PUSHED = Send. We don't know what will happen to the effect in this causal assignment. Sometimes the call goes through, and sometimes not. Because of this, we say that this response structure is indeterministic, and we separate deterministic from indeterministic causal relationships by separating deterministic vs. indeterministic response structures.

Deterministic Response Structure In a *deterministic response structure*, every causal assignment completely determines the value of the effect.

Indeterministic Response Structure In an *indeterministic response structure*, there is at least one causal assignment that does not determine the value of the effect.

3.1.2 Module 3 | An Abstract Example: Red and Blue Squares

Having translated the (simulated) cell phone example into the language of variables and response structures, we can now discuss indeterministic causation abstractly. We will use an example that has the same structure as the cell phones, but that involves abstract red and blue squares that have no connection to objects with which you are familiar.

Assume you have a pair of squares displayed on a computer. The state of the red square causes the state of the blue square. Suppose you can change the causal assignment for the red square from solid to hatched (dots), or vice versa, by clicking on it. The "Results Table" will record how many times the effect (blue square) came out solid or hatched in response to the causal assignment you gave to the red square: 1) red square = solid, 2) red square = hatched. Suppose you click on the red square enough times to bring about causal assignment 1) (solid red square) at least 10 times and causal assignment 2) (hatched red square) at least 10 times.

After experimenting, you find that by setting the red square to hatched, you can guarantee that the blue square will be hatched. Thus, making the state of the red square hatched fully determines the state of the blue square.

After experimenting, you also find that by setting the red square to solid, you cannot guarantee that the blue square will be solid or hatched. Thus, making the state of the red square solid does not fully determine the state of the blue square.

Since we don't know what will happen to the blue square in one of the causal assignments for the red square, the response structure is indeterministic. The red square is a cause of the blue square, but an indeterministic cause.

Compare the response structure for the colored square experiment and the cell phone.

RESPONSE STRUCTURE FOR THE COLORED SQUARE EXPERIMENT

Causal Assignment	Effect: Blue Square
Red Square = Solid	???
Red Square = Hatched	Hatched

RESPONSE STRUCTURE FOR THE CELL PHONE SYSTEM

Causal Assignment	Effect: Call Connected
BUTTON PUSHED = Send	???
BUTTON PUSHED = End	No

3.1.3 Module 3 | Individuals and Populations

So far, we have described indeterministic causation as a feature of a response structure, which applies to a particular individual or individuals. In the cell phone and red-blue-square examples, we were dealing with a single individual. We discovered that the causal relationship between placing a call (hitting “Send”) and having it go through is indeterministic by repeatedly placing the same phone in the same causal assignment and observing different effects at different times. Sometimes the call got through, and sometimes it didn’t. In the red-square blue-square example, we did the same thing. The experiment allowed us to repeat the same causal assignment many times for the same *individual*. When we assigned the red square to solid, sometimes we observed a solid blue square, and sometimes we observed a hatched blue square. Thus, we discovered that, for this individual, the response structure for the blue square is indeterministic.

As we stressed in the module on Variable Causation, scientists often cannot examine the response structure of particular individuals - they can only gather data on a population of individuals, each of whom are given one of many possible causal assignments. So first, what is it for causation in a population to be indeterministic?

Indeterministic Causation in a Population If there are two individuals in a population who exhibit different effects even though they have the same causal assignment, then there is *apparent indeterministic causation in the population*.

Recall the idea of response structure uniformity in a population:

Response Structure Uniformity A population has *response structure uniformity* for a given effect if every individual in the population has the same response structure for that effect.

Notice that this definition doesn’t say anything about whether each individual has a deterministic or indeterministic response structure, it only insists that they all have the *same* response structure.

We can tell if a population of individuals with deterministic response structures violates response structure uniformity if we give two individuals the same causal assignment but then observe them to have different effects. This won’t always happen, because two individuals that have different response structures might have the same response for all of their causal assignments but one, and if we don’t put them in the causal assignment that shows the difference, we might never observe that they are different.

So what happens to this problem if we don’t assume that every individual in a population has a deterministic response structure? What happens if we allow for indeterministic response structures?

In the Colored Square example above, you performed the same experiment over and over on the same pair of red and blue squares. Suppose you are given a population of 20 *distinct* individual pairs. For each of the individual pairs, you can click on the red square to change its causal assignment. You can change all the red squares *once*, and notice how many of the blue squares match. Then you repeat the experiment, and again notice how many of the blue squares change.

After experimenting, you observe that the causal assignment that you give to the red square does not have the same effect on every member of the population. There were almost certainly some pairs of red and blue squares that you gave the same causal assignment but that had different responses for the effect, thus the response structure is indeterministic.

Causation in a population can *appear* indeterministic for two reasons:

1. Some individual in the population has an indeterministic response structure.

2. Every individual in the population has a deterministic response structure, but the population is not response structure uniform.

If, in the experiment with the population of the 20 pairs of red and blue squares, we took the individual Colored Square from the last page and duplicated it 20 times, then the population would appear indeterministic for the first reason.

If, in the experiment with the population of the 20 pairs of red and blue squares, we had 20 different individual red-blue pairs, some of which had one deterministic response structure and some of which had another, then the population would appear indeterministic for the second reason.

If we took the first option, and created the population with 20 duplicates of the single pair example, how could you tell?

3.1.4 Module 3 | Cell Phones Revisited

Both the cell phone and Colored Square examples you used in the previous sections are instances of indeterministic causation. In this section we discuss how an indeterministic response structure only *appears* to be indeterministic because some of the causes for an effect were left out.

Suppose in the original example on the cell phone, you only managed to get about half of your calls to go through. Why? Isn't hitting the "SEND" button a cause of a call getting through? The answer is yes, it is a cause, but not the *only* cause.

We only showed you part of the story in the cell phone example. Here we uncover another cause of getting the call to go through: whether you are in range of a cell phone tower or not. You cannot control whether you are in range of the tower – but you may be able to observe it.

3.1.5 Module 3 | Red and Blue Squares Revisited

In this section we again abstract away from cell phones to colored squares to secure the idea of how causation can appear indeterministic if some causes are unobserved. We now *add a square* to the red and blue squares we used previously.

What happened in the cell phones is the same as what happened in the red and blue square simulation: we only gave you part of the story. In the following example you get the full picture.

Now, you can intervene to change the state of the red square, but the computer has control over a newly added green square. The causal assignment for the states of the red and green square is created partly by you clicking on the red square and partly by the computer choosing the state of the green square. When you change the state of the red square, the program assigns a value at random to the green square. Sometimes it assigns it hatched green and sometimes solid green. The red and green squares are both causes of the blue square.

Like the cell phone situation, once you know about the hidden factor (the state of the green square in this case and whether the car was in range of the cell tower in the cell phone case), you see that the causal assignment (of both the red and green squares) fully determines the effect. Because of the way it has been set up, *you* have no control over the green square – only the red one. Nevertheless, the causal assignment among the red and green square totally determines the state of the blue square, which is the effect in this system. In this *expanded* system, the effect now has a deterministic response structure.

This table summarizes the response structure for the blue square in the expanded system, but leaves one response blank for you to complete:

RESPONSE STRUCTURE FOR THE SYSTEM OF SQUARES

Causal Assignment	Causal Factor 1: RED	Causal Factor 2: GREEN	Effect: BLUE
1	Solid	Solid	Solid
2	Solid	Hatched	Hatched
3	Hatched	Solid	
4	Hatched	Hatched	Hatched

In completing the table above, it may help to remind yourself that the example that included no green square is set up identically to the one here, except that you are not *shown* the green square. In the previous example, it was a *hidden variable*. This is exactly the same as the cell phone example. In both cases, the underlying models are identical, but in one you are shown the range from the tower, and in the other it is kept hidden from your view.

3.2 Indeterminism and Pseudo-Indeterminism

3.2.1 Module 3 | Introduction

When the response structure for the cell phones includes BUTTON PUSHED and TOWER RANGE, i.e., *all* the causes of whether the call goes through, it is a deterministic response structure. When it includes only BUTTON PUSHED, it is an indeterministic response structure. Since the indeterminism in the second case is the result of leaving out a variable, we call it *pseudo-indeterminism*. In this section, we make the idea of pseudo-indeterminism a little more precise, contrast it with true indeterminism, and examine the practical consequences of the distinction.

3.2.2 Module 3 | Pseudo-Indeterminism

When the response structure for the cell phones includes all the causes of the call going through: BUTTON PUSHED and TOWER RANGE, it is deterministic. When it includes only BUTTON PUSHED, it is indeterministic. Since placing a call on the cell phone only appears to be an indeterministic cause of the call getting through when TOWER RANGE is left out of the story, we label the relationship pseudo-indeterministic. Here we explain pseudo-indeterminism and make it a little more precise.

Consider the case of malaria in the context of deterministic vs. indeterministic causation. Leaving out the irrelevant factor of whether the individual drank gin and tonics regularly, we represented the causal structure governing whether or not Sam (or others) would get malaria with the following response structure:

RESPONSE STRUCTURE FOR THE MALARIA CASE				
Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Causal Factor 3: INOCULATED	Effect: MALARIA
1	True	True	True	False
2	True	True	False	False
3	True	False	True	False
4	True	False	False	True
5	False	True	True	False
6	False	True	False	False
7	False	False	True	False
8	False	False	False	False

Since each of the eight possible causal assignments in the table are different with respect to the variables: BITTEN, HAS GENE, and INOCULATED, and only one possible outcome for the effect is listed for each assignment, the effect has a deterministic response structure.

Suppose, however, that the system was again the same, but we had no knowledge of the genetic make-up of each individual, e.g., Table 3200-2 (below).

RESPONSE STRUCTURE FOR MALARIA – WITH GENE UNOBSERVED				
Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Causal Factor 3: INOCULATED	Effect: MALARIA
1	True	Unobserved	True	False
2	True	Unobserved	False	False
3	True	Unobserved	True	False
4	True	Unobserved	False	True
5	False	Unobserved	True	False
6	False	Unobserved	False	False
7	False	Unobserved	True	False

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RESPONSE STRUCTURE FOR MALARIA – WITH GENE UNOBSERVED (Continued)

Causal Assignment	Causal Factor 1: BITTEN	Causal Factor 2: HAS GENE	Causal Factor 3: INOCULATED	Effect: MALARIA
8	False	Unobserved	False	False

Suppose that we only observed whether each person was bitten by a malaria carrying mosquito and whether each person was inoculated. Even though there are in fact eight different possible causal assignments, observationally we can now only distinguish four. In Table 3200-2, before looking at the effect, assignments 1 and 3 now appear identical to us, as do assignments 2 and 4, 5 and 7, and 6 and 8.

We say that the response structure for malaria in Table 3000-2 is pseudo-indeterministic.

Pseudo-indeterministic Response Structure A response structure for an effect E is *pseudo-indeterministic* when i) it is an indeterministic response structure for E, and ii) there are other causes of E that, when added, make the response structure deterministic.

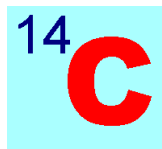
In a pseudo-indeterministic system, the indeterminism arises only because we have not observed all the relevant causal factors. If we had, then the system would appear deterministic. So even if we believe that the world is at bedrock deterministic, whether it appears deterministic is relative to what we are or are not able to observe.



Until the 20th century, most scientists believed that the world was fundamentally deterministic. Pierre Simon de Laplace, one of the greatest French mathematicians and astronomers of the 18th century, claimed that if we could know the current state of the universe down to the smallest detail, we could predict exactly how it will be at any time in the future and how it was at any time in the past. He was well aware that most things *appeared* indeterministic, but he was convinced that it was the limitations on what we could observe and measure and not the nature of the world that gives rise to the indeterminism we observe. Laplace believed that we live in a pseudo-indeterministic world.

3.2.3 Module 3 | True Indeterminism

In this century, the theory of Quantum Mechanics, which describes the world at very small scales (an atom and smaller), has convinced most scientists that Laplace was wrong – the world is not pseudo-indeterministic. In the standard interpretation of Quantum Mechanics, the world is *truly indeterministic*, especially at the sub-atomic scale.



For example, the isotope Carbon 14 is radioactive. Carbon atoms that have a total of 14 protons and neutrons in their nucleus sometimes radioactively decay. If we had a trillion Carbon 14 atoms in a box, only a small percentage of them would decay within an hour, and nothing in physical theory can tell us which atoms will decay in that time and which will not. A disciple of Laplace might argue that the indeterminism among Carbon 14 radioactive decay is only in appearance – if we just knew more about each Carbon 14, i.e., other causes of radioactive decay, then we could indeed predict perfectly whether it would or would not decay in the first hour. Most modern physicists disagree – they believe that there is nothing else about the atoms we could learn that would help in predicting whether or not a Carbon 14 atom decays in any specific time period. They believe that radioactive decay is a truly indeterministic phenomenon.

There remain to this day a minority of physicists that refuse to believe that “God plays dice” with the universe, and contend that we simply don’t understand the “hidden” variables in cases such as atomic decay. These scientists pursue “hidden variable theories” that try to describe the atomic world as pseudo-indeterministic, not truly indeterministic.

Truly Indeterministic Response Structure A response structure for an effect E is *truly-indeterministic* when i) it is an indeterministic response structure for E, and ii) there is no set of causes we can add that will make it a deterministic response structure.

3.2.4 Module 3 | The Practical Situation

As you can see, whether or not a response structure for an effect is indeterministic is sometimes a question of what we observe. If a system appears indeterministic, it might be because it is pseudo-indeterministic and we have left out some important causes in our description of the response structure, or it might be because the system is truly indeterministic.

For virtually any effect that we care to study in Biology, Psychology, or any Social Science, we are not even close to being able to identify and measure *all* of its causes. Thus the distinction between pseudo-indeterministic and truly indeterministic systems turns out to be more of philosophical than practical interest.

What are the causes of depression? Brain chemistry, emotional trauma in childhood, poor relationships, death of a loved one? Perhaps, but surely the response structure for depression with these variables is not deterministic. Adding other causes besides these might help us predict depression more reliably, but we are unlikely to ever be in a position to predict or prevent depression with perfect certainty. We are unlikely to ever be able to even identify a fraction of all the causes of depression. The same is true of conflict, crime, and economic health. Given this, causal reasoning in any domain is by necessity about indeterministic relationships. Whether they are pseudo-indeterministic or truly indeterministic matters little. What matters is how to understand and reason about indeterministic causal relationships.

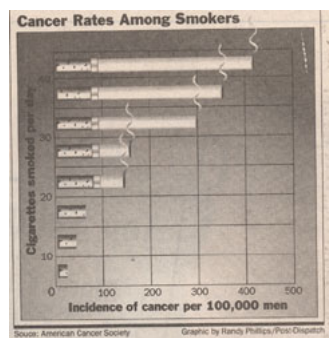
3.3 Indeterministic Causation

3.3.1 Module 3 | Introduction

Having equipped ourselves with plenty of terminology, we now need to include indeterminism as part of an account of causation among variables.



What do we mean when we say that smoking causes lung cancer, even though a causal assignment of smoking or of not smoking does not determine the outcome for lung cancer?



Roughly, we mean that the chances of getting lung cancer would be different among people *assigned* to smoke than among people assigned not to smoke. Is this claim proved by evidence that shows that people who *choose* to smoke more have a greater chance of getting lung cancer, like the chart above? No, because there might be a common cause of choosing to smoke and getting lung cancer. Not likely, but possible. The chart shows us that although no amount of smoking guarantees lung cancer, more smoking is accompanied by a higher chance of getting lung cancer, that is, there is a statistical association between smoking and lung cancer. Such an association might be the result of an indeterministic causal relationship between smoking and lung cancer. In this section we give a basic account of indeterministic causation among variables, generalizing the account we gave in the previous module to include indeterminism.

3.3.2 Module 3 | Cell Phones and Indeterminism

Consider the causal system for the cell phone again.

In a response structure for CALL CONNECTED in the system in which the only cause under consideration is BUTTON PUSHED, there are exactly two causal assignments, one for each value we might give to BUTTON PUSHED.

RESPONSE STRUCTURE FOR THE SMALL CELL PHONE SYSTEM		
Causal Assignment	BUTTON PUSHED	Effect: CALL CONNECTED
1	Send	???
2	End	No

The problem is that we don't know what value to write in for the effect CALL CONNECTED when the causal assignment is BUTTON PUSHED = Send.

When we considered the full Cell Phone system, however, we included the variable: IN TOWER RANGE.

In this full version of the system the Response Structure has four different causal assignments, and is deterministic.

RESPONSE STRUCTURE FOR THE FULL CELL PHONE SYSTEM			
Causal Assignment	BUTTON PUSHED	IN TOWER RANGE	Effect: CALL CONNECTED
1	Send	Yes	Yes
2	Send	No	No
3	End	Yes	No
4	End	No	No

The reason the response structure for the *small* system appears indeterministic is that in the causal assignment labelled: BUTTON PUSHED=Send, we don't know whether the hidden variable IN TOWER RANGE was assigned the value Yes or No. We represented this uncertainty by putting "???" in for the value of the effect CALL CONNECTED.

We can often do better at representing our uncertainty than just writing "???", however. In all the Cell Phone experiment trials, we could have a computer set the value of IN TOWER RANGE by flipping a computer's version of a fair coin. About 50% of the time, IN TOWER RANGE should equal Yes, and about 50% of the time it should equal No. That means that about 50% of the time you pushed the Send button, the call went through, and 50% it didn't. If you only did a few trials, you might have gotten a proportion quite different from 50% by luck, but if you did a lot of trials it's very unlikely you got something very different from 50%. In any event, instead of putting "???" into the response structure for the small system, we could have put something more informative about our uncertainty, like: Yes = 50%, No = 50%.

RESPONSE STRUCTURE FOR THE SMALL CELL PHONE SYSTEM		
Causal Assignment	BUTTON PUSHED	Effect: CALL CONNECTED
1	Send	Yes = 50%, No = 50%
2	End	No

In our Cell Phone simulations, although pushing the Send button doesn't guarantee that your call will go through, it does change the chances of the call getting through. If you don't push the Send button, the chances of the call getting through are 0. If you do push it, they change to 50%. Thus pushing the send button is a cause of the call going through.

3.3.3 Module 3 | Quantifying Indeterministic Causation

Although we must accept that some response structures are indeterministic, we don't have to accept ones in which there are completely vague entries for the effect like "???". We can at least try to quantify our uncertainty about whether the effect will occur in each causal assignment. We do so with probabilities, or chances.

Example (A) Quantifying Indeterminism: Smoking and Lung Cancer

For example, consider the following (fictitious) indeterministic response structure for smoking and lung cancer.

RESPONSE STRUCTURE FOR SMOKING AND LUNG CANCER			
Causal Assignment	Cause	Effect	
	SMOKER	LUNG CANCER = Yes	LUNG CANCER = No
1	Yes	15%	85%
2	No	1%	99%

Instead of a standard response structure table, in which there is only one column for the effect, we list, for each causal assignment, the chances the effect will be cancer and the chances it will be not cancer. In this example, we don't know with certainty what the effect will be in any causal assignment, but using a table like this we can express how the *chances* of the effect change as we change the causal assignment.

If the effect is a variable that can take on more than two values, then the response structure must contain a probability for each value, and the probabilities must add to 1.0.

Example (B) Quantifying Indeterminism: Education and Income

For example, suppose that we want to represent how the variable INCOME depends on the variable EDUCATION. Suppose that EDUCATION can take on the values High School, College, or Graduate Degree, and INCOME can take on the values Low, Medium, or High. Below is an example of an indeterministic response structure in which we give the chances for achieving different levels of INCOME as a function of the causal assignment for EDUCATION.

RESPONSE STRUCTURE FOR EDUCATION AND INCOME				
Causal Assignment	Cause	Effect		
	EDUCATION	INCOME = Low	INCOME = Medium	INCOME = High
1	High School	60%	35%	5%
2	College	20%	70%	10%
3	Grad Degree	5%	70%	25%

According to this table, the chances of earning a sizable amount of money increase if we increase the amount of education in the causal assignment. The probabilities for the values of the effect must add to 1, and in general the way the total of 1 is divided up among the different values is called a “probability distribution.”

In general, if we can't construct a deterministic response structure for an effect variable, we want a response structure in which we list the probability distribution over the values the effect might take on. This gives us a quantitative sense of how the chances of the effect depend on the causal assignment given.

3.4 Defining Causation with Indeterminism

3.4.1 Module 3 | Direct Causation with Indeterminism

We now have all the pieces we need to include indeterminism in an account of causation among variables. In this section we expand the definition of direct causation to include indeterminism, and in the next section we distinguish between direct and indirect causation.

Recall the two key definitions we gave in the module on Variable Causation.

Test Pair of Causal Assignments for X Two causal assignments C1 and C2 are a *test pair of causal assignments* for X if and only if they are identical except for the values assigned to variable X

Direct Cause In a system of variables S , X is a *direct cause* of Y relative to S if and only if there is at least one test pair of causal assignments for X in the response structure for Y across which Y differs.

The idea behind test pairs of causal assignments is to keep everything in the system the same except for one variable, which we make different between one causal assignment and another. If the effect changes, then we have causation.

What is it for the effect to *change*, however? Does it mean that the effect must take on a different value in the two causal assignments? Or is it enough that the *chances* for the effect change? Clearly, in the case of indeterministic causation, only the chances need to change. So let's modify the definition of cause ever so slightly to accomodate indeterminism:

Direct (Indeterministic) Cause In a system of variables S , X is a *direct cause* of Y relative to S if and only if there is at least one test pair of causal assignments for X in the response structure Y across which the probability distribution over Y differs.

The “probability distribution” means, roughly, the probability given to each of the possible values Y can take on. If the probability given to any of Y 's values changes, then the probability distribution over Y has changed.

Example (C) Probability Distributions: Socioeconomic Class

For example, consider three different probability distributions over a variable Socioeconomic Class, which we will abbreviate SES, which can take on three different values: Low, Middle, High. We write $p(\text{SES} = \text{Low})$ to refer to the probability of having low socioeconomic status.

Probability Distributions over Socioeconomic Class			
Probability Distribution	$p(\text{SES} = \text{Low})$	$p(\text{SES} = \text{Middle})$	$p(\text{SES} = \text{High})$
1	0.50	0.30	0.20
2	0.35	0.30	0.35
3	0.30	0.30	0.40

In Row 1, according to the first probability distribution, the chances of having low socioeconomic status are 50%, and the chances of having high socioeconomic status are 20%. Notice that although all three distributions are different, they all agree on the probability of having middle socioeconomic status being 30%. If two probability distributions assign different probabilities to *any value*, they are different.

In general, if the probability distribution over an effect Y changes at all across a test pair for X , then X is an indeterministic cause of Y .

Just as we argued in the previous module that the effect need not be different across *every* test pair when we defined direct cause, the parallel holds for our definition of direct indeterministic cause. The chances, or probability distribution over the effect don't have to change across every test pair of causal assignments, only across at least one pair.

Example (D) Indeterministic Causation: Income and Happiness

For example, consider an indeterministic response structure that depicts the effect of INCOME on HAPPINESS.

RESPONSE STRUCTURE FOR INCOME AND HAPPINESS			
Causal Assignment	Cause	Effect	
	INCOME	$p(\text{HAPPINESS} = \text{Low})$	$p(\text{HAPPINESS} = \text{High})$
1	Low	60%	40%
2	Medium	45%	55%
3	High	45%	55%

According to this response structure, changing INCOME from Low to Medium improves the chances for high HAPPINESS, but changing from Medium to High INCOME doesn't have any effect. After you earn more than

low income, extra happiness depends more on other factors.

3.4.2 Module 3 | Direct vs. Indirect Causation

Roughly, if X is a direct cause of Y relative to some causal system S , then we can hold the other variables in S constant and produce a change in Y as a result of changing X . If X is only an indirect cause of Y , however, then all of X 's influence on Y must go through some other variable in S , say V . X must first produce a change in V , which must in turn produce a change in Y . In the response structure for Y , however, any test pair for X will involve two assignments in which V does *not* change, and so X will not appear as a direct cause of Y . That does *not* mean that X is not a *cause* of Y , it just means that it is not a *direct* cause of Y .

Example (E) Indirect Causation: Chicken Pox



Consider how children get Chicken Pox. Exposure to another child who has the disease causes viral infection, and viral infection causes the chicken pox rash. Exposure to another child is an *indirect* cause of the rash, but not a *direct* cause. What is the difference between a direct or indirect cause, and how can our theory handle the difference?

Lets work this through for the Chicken Pox example. Suppose we represent the system with thre variables: Exposed: [yes, no], Infected: [yes, no], Rash: [yes, no]. Suppose the response structure for Rash is as follows.

RESPONSE STRUCTURE FOR Chicken Pox Rash				
Causal Assignment	Cause 1	Cause 2	Effect	Effect
	Exposed	Infected	p(Rash = yes)	p(Rash = no)
1	Yes	Yes	0.70	0.30
2	Yes	No	0.0	1.0
3	No	Yes	0.70	0.30
4	No	No	0.0	1.0

What are the test pairs for Exposed in this response structure? One is assignments 1 and 3, and the other is assignments 2 and 4. In assignments 1 and 3, we have set Infected to yes. The fact that in one case we assigned Exposure to yes and in the other to no will make no difference to whether a child gets the rash. Likewise in assignments 2 and 4, where we have assigned Infected to no. Thus, appropriately, this response structure indicates that Exposure is *not* a direct cause of Rash.

By common sense, of course, Exposure *is* a cause of getting the Chicken Pox rash. Does anything in the response structure for Rash in the system that includes Exposed and Infected tell us that Exposure is an indirect cause? No. All the response structure for Rash tell us is which of the potential causes are direct causes of Rash.

Intuitively, when we ask if one variable X is a *cause* of another Y , we don't care if X is a direct or an indirect cause of Y . We only really want to know whether we can change Y by intervening to change X . We really want to know the response structure for Y in the system involving *only* X and Y . X is a cause of Y if X is a direct cause of Y in the system that involves *only* X and Y .

Example (F) Testing for Indirect Causation: Chicken Pox

In the Chicken Pox example, asking whether the Chicken Pox rash is contagious, that is, whether Exposure to another child who has Chicken Pox changes the chances of getting the Chicken Pox rash means asking whether Exposure is a direct cause of Rash in the system involving only Exposure and Rash.

Response Structure For Chicken Pox Rash in Reduced System

Causal Assignment	Cause Exposed	Effect p(Rash = yes)	Effect p(Rash = no)
1	Yes	0.50	0.50
2	No	0.01	0.99

In this reduced system, we need only assess the chances of getting the Chicken Pox rash in two causal assignments, exposed to another child or not. Since not all cases of exposure result in infection, we put the chances of getting a rash at 50% for kids who have been exposed, and the chances for catching it without being exposed to another kid at 1%.

Thus, our definition for when one variable is a *cause* of another is as follows.

Cause X is a *cause* of Y if and only if, in the system of variables involving only X and Y, there is at least one test pair of causal assignments for X in the response structure for Y across which the probability distribution over Y differs.

A more compact but equivalent way to state this definition is:

Cause X is a *cause* of Y if and only if, in the system of variables involving only X and Y, X is a direct cause of Y.

3.5 Summary

3.5.1 Module 3 | Module Summary

Having worked through the module, how might we use its basic ideas to answer Mrs. Smith, who doesn't want to pay for removing the Taylor burners that have asbestos because she says: "My relatives lived in houses with these heaters for 40 years, and none of *them* got cancer"?

Perhaps we might respond as so: "Asbestos exposure need not guarantee cancer to be a cause of it. It need only increase the chances of getting cancer, and scientists believe it does precisely that. Your relatives were lucky to avoid cancer, even though they increased their chances of getting it by living with Taylor burners. I don't want to expose my own children to asbestos and increase their risk of getting cancer, even if it is possible that they will be as lucky as your relatives, so please remove them before we move in." Flustered, Mrs. Smith sees the irrefutable logic in your answer and agrees immediately to all of your demands. You promptly send a huge donation to the authors of Causal and Statistical Reasoning, and all are happy. Only kidding about the donation.

This module covered the following six concepts:

3.5.2 Concepts Covered in Module 3

1. Indeterministic Response Structures
2. Sources of Indeterminism
3. Pseudo-indeterminism vs. True Indeterminism
4. Quantifying Indeterministic Causation
5. Definition of Indeterministic Causation
6. Direct vs. Indirect Causation

The *response structure* for an effect describes the effect in every possible causal assignment of the causes mentioned in the system. In a *deterministic response structure*, every causal assignment completely determines the value of the effect.

In an *indeterministic response structure*, there is at least one causal assignment that does not determine the value of the effect.

For many effects of interest, like cancer, a response structure that includes causal assignments for all the causes we know about now still *appears indeterministic*. Someday in the future, we will perhaps actually be able to measure *all* the causes of cancer. Suppose that when we do, it turns out the full response structure for cancer turns out to be deterministic. In that case the current response structure for cancer is *pseudo-indeterministic*. The response structure we currently know about for cancer is indeterministic, but it is embedded within a bigger response structure that is deterministic.

A response structure for an effect E is *pseudo-indeterministic* when:

- it is an indeterministic response structure for E, and
- there are other causes of E that, when added, make the response structure deterministic.

Unfortunately, the system that we specify and observe in practice is rarely more than a partial version of the causal system that governs the world. For almost no medical or social or psychological system, for example, do we believe we can identify and observe all the causal factors that contribute to producing an effect. For example, although the Surgeon General concluded in 1965 that smoking causes lung cancer, he did not conclude (and it is not true) that all smokers get lung cancer. Even considering dietary habits, family history, and other environmental and life-style factors, we cannot identify any causal assignment that is sufficient to determine whether someone will or will not get lung cancer. We can identify many situations that result in lung cancer more frequently, but none that guarantee it, and none that guarantee preventing it.

Will it ever be possible to specify all the causes of lung cancer, so that we can build a deterministic response structure? No one knows. If it is not possible, then the response structure for cancer is truly indeterministic. A response structure for an effect E is *truly-indeterministic* when:

- it is an indeterministic response structure for E, and
- there is no set of causes we can add that will make it a deterministic response structure.

Whether or not the world is fundamentally deterministic or indeterministic, in practice we can only specify and measure some of the relevant factors that cause an effect of interest. Indeed, if we don't make prior decisions about which factors might be relevant, the number of potential causal assignments is practically infinite. In almost every case, however, no matter how many factors we include in the causal system we *specify*, we will leave out some that are relevant. As a result, in practice our representations of causal structures are always partial, and we have to accept the fact that indeterministic response structures are the rule, not the exception.

To deal with indeterministic response structures, we quantify our uncertainty about the effect for each assignment we might give to the causes. In a quantified indeterministic response structure, for each causal assignment we give the chances for each possible value of the effect variable, that is, we give the probability distribution over the effect.

One variable X is a direct cause of another Y relative to some system S just in case in the response structure for Y there are two causal assignments that are identical except for the value of X, that is, if there is a *test pair* of causal assignments for X, such that the probability distribution over Y is different across the two assignments.

One variable X is a cause, direct or indirect of another Y just in case, in the system involving just X and Y, X is a direct cause of Y.